



# **CS 412 Intro. to Data Mining**

## **Chapter 3. Data Preprocessing**

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# Chapter 3: Data Preprocessing

- ## Summary



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# What is Data Preprocessing? — Major Tasks

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## ☐ Data cleaning

- ☐ Handle missing data, smooth noisy data, identify or remove outliers, and resolve inconsistencies

## ☐ Data integration

- ☐ Integration of multiple databases, data cubes, or files

## ☐ Data reduction

- ☐ Dimensionality reduction
- ☐ Numerosity reduction
- ☐ Data compression

## ☐ Data transformation and data discretization

- ☐ Normalization
- ☐ Concept hierarchy generation

# Why Preprocess the Data? — Data Quality Issues

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- ❑ Measures for data quality: A multidimensional view
  - ❑ Accuracy: correct or wrong, accurate or not
  - ❑ Completeness: not recorded, unavailable, ...
  - ❑ Consistency: some modified but some not, dangling, ...
  - ❑ Timeliness: timely update?
  - ❑ Believability: how trustable the data are correct?
  - ❑ Interpretability: how easily the data can be understood?

Handwritten notes in blue ink:

- Dimension
- Handwritten Burmese text: ကိုယ်လုပ်ဆောင်နေတာ
- Handwritten Burmese text: အမှန်တရား

# Chapter 3: Data Preprocessing

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- ❑ Data Preprocessing: An Overview
- ❑ Data Cleaning or Data Cleansing
- ❑ Data Integration
- ❑ Data Reduction and Transformation
- ❑ Dimensionality Reduction
- ❑ Summary



all DATA

# Data Cleaning

ဒီတော့ဒီစာဖတ်ပြီး မိရတာက မှန်လို့မဟုတ်ဘူး

- ❑ Data in the Real World Is Dirty: Lots of potentially incorrect data, e.g., instrument faulty, human or computer error, and transmission error
- ❑ Incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
  - ❑ e.g., *Occupation* = " " (missing data)
- ❑ Noisy: containing noise, errors, or outliers
  - ❑ e.g., *Salary* = "-10" (an error)
- ❑ Inconsistent: containing discrepancies in codes or names, e.g.,
  - ❑ *Age* = "42", *Birthday* = "03/07/2010"
  - ❑ Was rating "1, 2, 3", now rating "A, B, C"
  - ❑ discrepancy between duplicate records
- ❑ Intentional (e.g., *disguised missing data*)
  - ❑ Jan. 1 as everyone's birthday?

missing DATA

DATA မှီ

# Incomplete (Missing) Data

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- ❑ Data is not always available
  - ❑ E.g., many tuples have no recorded value for several attributes, such as customer income in sales data
- ❑ Missing data may be due to
  - ❑ Equipment malfunction
  - ❑ Inconsistent with other recorded data and thus deleted
  - ❑ Data were not entered due to misunderstanding
  - ❑ Certain data may not be considered important at the time of entry
  - ❑ Did not register history or changes of the data
- ❑ Missing data may need to be inferred



# How to Handle Missing Data?

- ❑ Ignore the tuple: usually done when class label is missing (when doing classification)—not effective when the % of missing values per attribute varies considerably
- ❑ Fill in the missing value manually: tedious + infeasible? *การใส่ค่าใน DATA ของเรา - ไม่ค่อยดี*
- ❑ Fill in it automatically with *missing value ใน DATA*
  - ❑ a global constant : e.g., “unknown”, a new class?!
  - ❑ the attribute mean *mean ของ column*
  - ❑ the attribute mean for all samples belonging to the same class: smarter
  - ❑ **the most probable value: inference-based such as Bayesian formula or decision tree**

# Noisy Data

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- ❑ **Noise:** random error or variance in a measured variable
- ❑ **Incorrect attribute values** may be due to
  - ❑ Faulty data collection instruments
  - ❑ Data entry problems
  - ❑ Data transmission problems
  - ❑ Technology limitation
  - ❑ Inconsistency in naming convention
- ❑ **Other data problems**
  - ❑ Duplicate records
  - ❑ Incomplete data
  - ❑ Inconsistent data

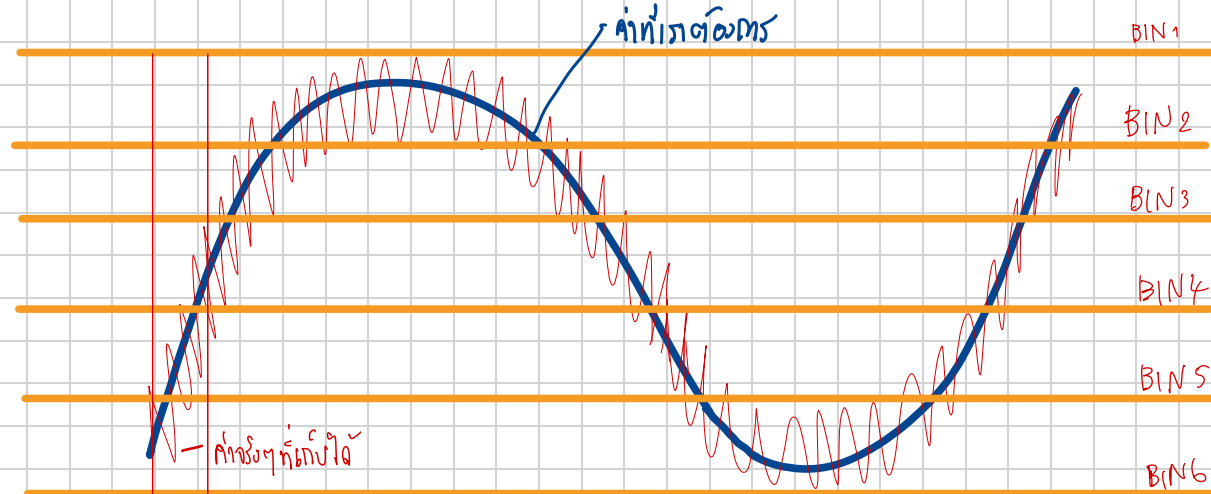
# ✓ How to Handle Noisy Data?

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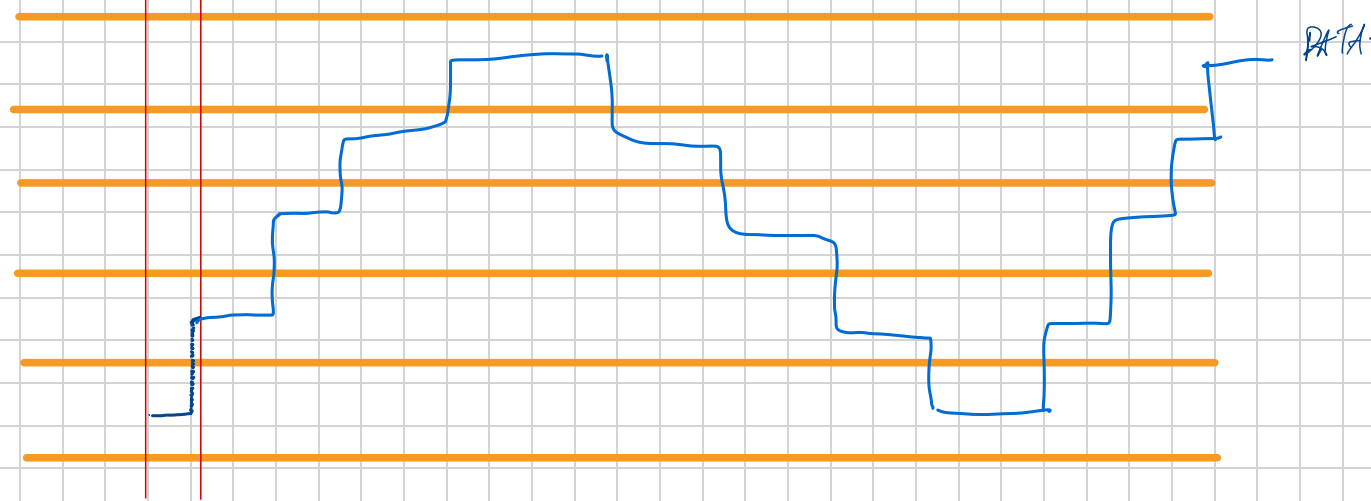
- ❑ Binning
  - ❑ First sort data and partition into (equal-frequency) bins
  - ❑ Then one can **smooth by bin means, smooth by bin median, smooth by bin boundaries**, etc.
- ❑ Regression
  - ❑ Smooth by fitting the data into regression functions
- ❑ Clustering
  - ❑ Detect and remove outliers
- ❑ Semi-supervised: Combined computer and human inspection
  - ❑ Detect suspicious values and check by human (e.g., deal with possible outliers)



# วิธี BINNING



✓



# Data Cleaning as a Process

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## ❑ Data discrepancy detection

- ❑ Use metadata (e.g., domain, range, dependency, distribution)
- ❑ Check field overloading
- ❑ Check uniqueness rule, consecutive rule and null rule
- ❑ Use commercial tools
  - ❑ Data scrubbing: use simple domain knowledge (e.g., postal code, spell-check) to detect errors and make corrections
  - ❑ Data auditing: by analyzing data to discover rules and relationship to detect violators (e.g., correlation and clustering to find outliers)

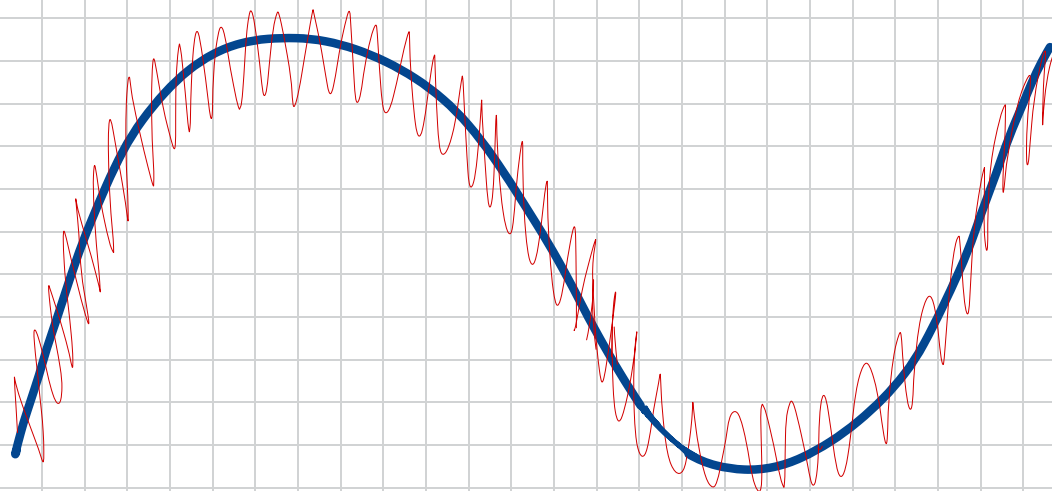
## ❑ Data migration and integration

- ❑ Data migration tools: allow transformations to be specified
  - ❑ ETL (Extraction/Transformation/Loading) tools: allow users to specify transformations through a graphical user interface
- ## ❑ Integration of the two processes
- ❑ Iterative and interactive (e.g., Potter's Wheels)

# Chapter 3: Data Preprocessing

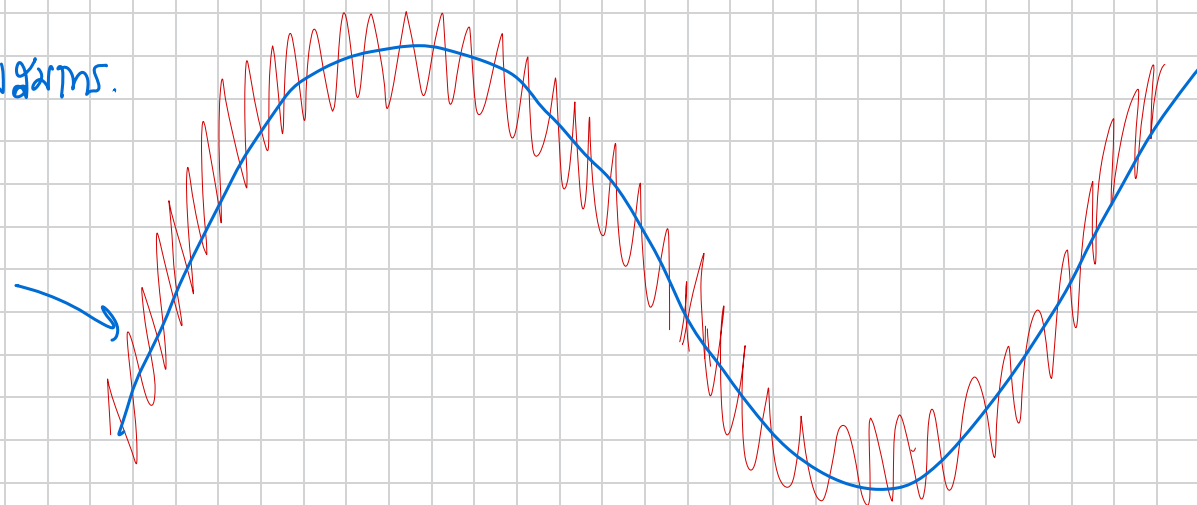
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- ❑ Data Preprocessing: An Overview
- ❑ Data Cleaning
- ❑ Data Integration 
- ❑ Data Reduction and Transformation
- ❑ Dimensionality Reduction
- ❑ Summary



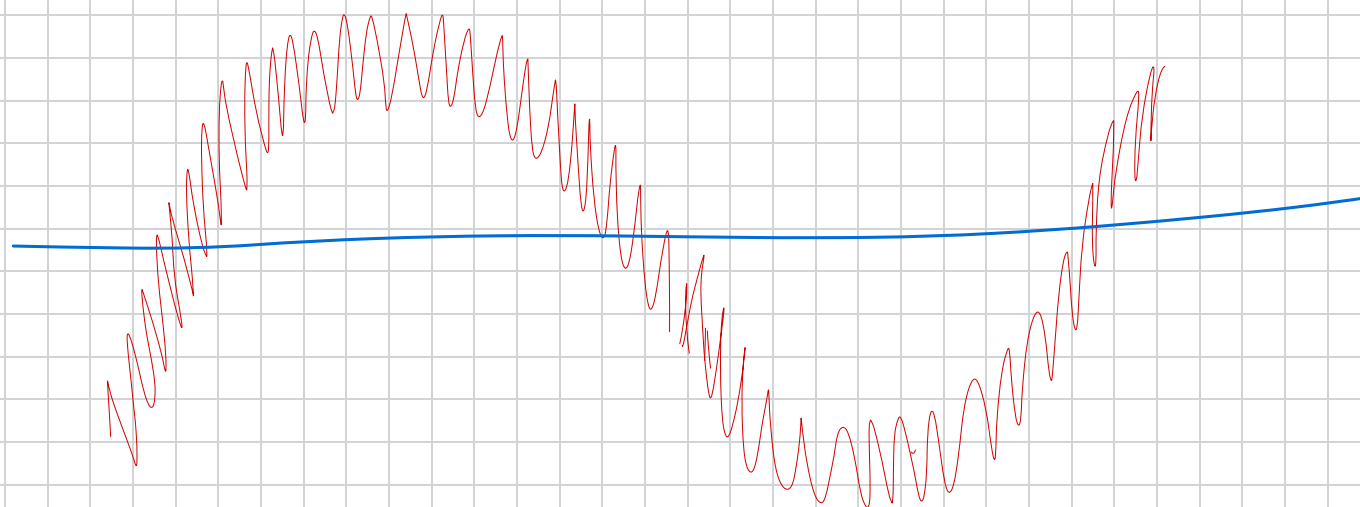
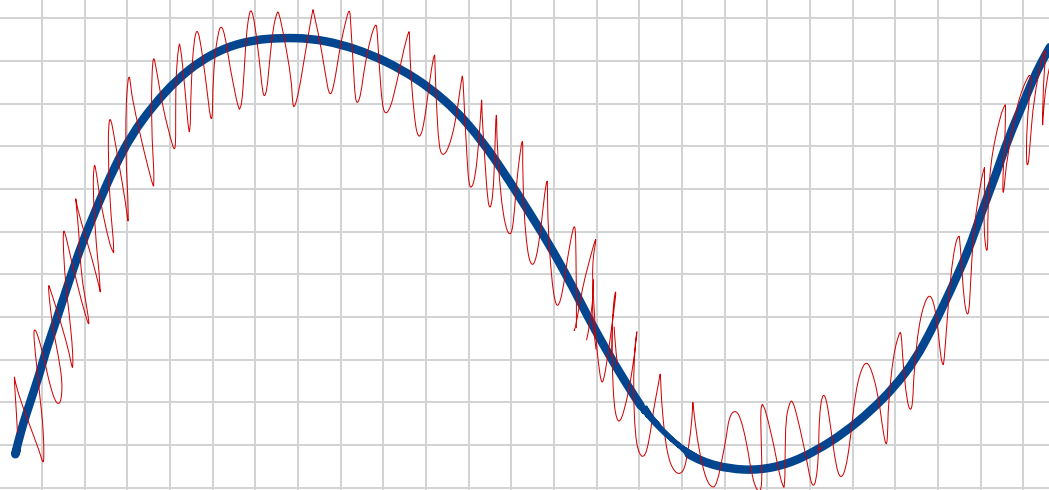
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Regression



# Data Integration

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- ❑ Data integration
  - ❑ Combining data from multiple sources into a coherent store
- ❑ Schema integration: e.g.,  $A.cust-id \equiv B.cust-\#$ 
  - ❑ Integrate metadata from different sources
- ❑ **Entity identification:**
  - ❑ Identify real world entities from multiple data sources, e.g., Bill Clinton = William Clinton
- ❑ Detecting and resolving data value conflicts
  - ❑ For the same real world entity, attribute values from different sources are different
  - ❑ Possible reasons: different representations, different scales, e.g., metric vs. British units

# Handling Redundancy in Data Integration

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- ❑ Redundant data occur often when integration of multiple databases
  - ❑ *Object identification*: The same attribute or object may have different names in different databases
  - ❑ *Derivable data*: One attribute may be a “derived” attribute in another table, e.g., annual revenue
- ❑ **Redundant attributes may be able to be detected by *correlation analysis* and *covariance analysis***
- ❑ Careful integration of the data from multiple sources may help reduce/avoid redundancies and inconsistencies and improve mining speed and quality

# Dimensionality Reduction Techniques

- ❑ **Feature selection:** Find a subset of the original variables (or features, attributes)
- ❑ **Feature extraction:** Transform the data in the high-dimensional space to a space of fewer dimensions

- Some typical dimensionality methods

שאלות פתוחות, מילוליות

ลดมิติข้อมูล

# Dimensionality Reduction

มิติที่ 1/2: ให้ความหมายของข้อมูลน้อยลง

## ❑ Curse of dimensionality

- ❑ When dimensionality increases, data becomes increasingly sparse
- ❑ Density and distance between points, which is critical to clustering, outlier analysis, becomes less meaningful
- ❑ The possible combinations of subspaces will grow exponentially

↓ ความหนาแน่น ↓ ระยะห่าง

มี DATA สั้นๆ คือเห็นภาพง่ายขึ้น

## ❑ Dimensionality reduction

- ❑ Reducing the number of random variables under consideration, via obtaining a set of principal variables

## ❑ Advantages of dimensionality reduction

- ❑ Avoid the curse of dimensionality
- ❑ Help eliminate irrelevant features and reduce noise
- ❑ Reduce time and space required in data mining
- ❑ Allow easier visualization



# Dimensionality Reduction Techniques

## □ Dimensionality reduction methodologies

2 วิธี.  
□ **Feature selection:** Find a subset of the original variables (or features, attributes)

□ **Feature extraction:** Transform the data in the high-dimensional space to a space of fewer dimensions

## □ Some typical dimensionality methods

□ Principal Component Analysis

□ Supervised and nonlinear techniques

□ Feature subset selection

□ Feature creation

6 ข้อ

พิจารณาข้อ

เลือกที่ใกล้เคียง

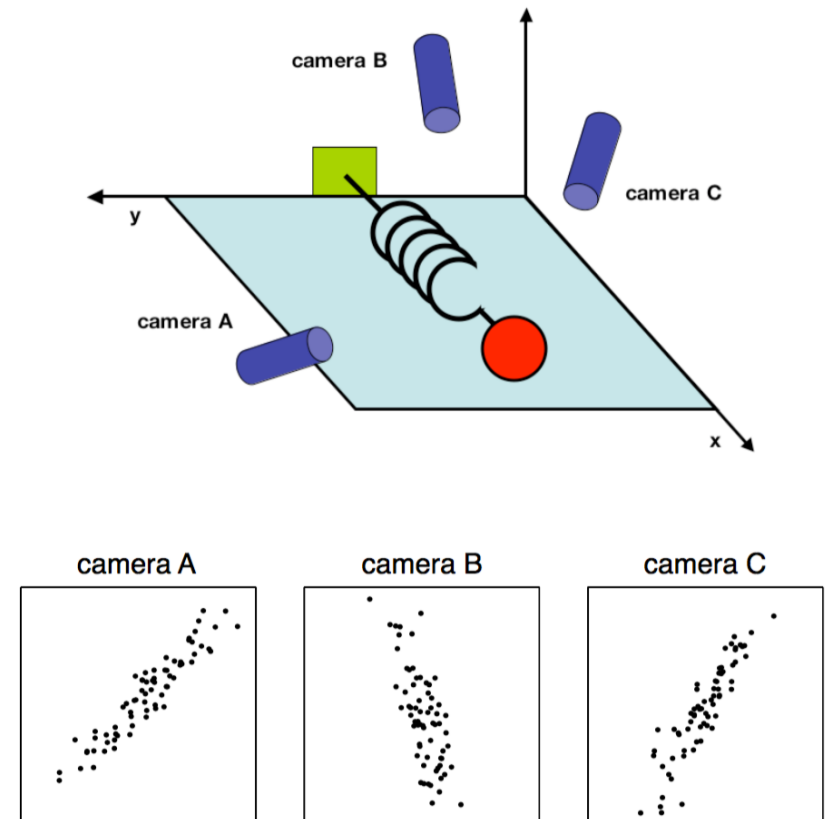
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~ Regression.

PCA

# Principal Component Analysis (PCA)

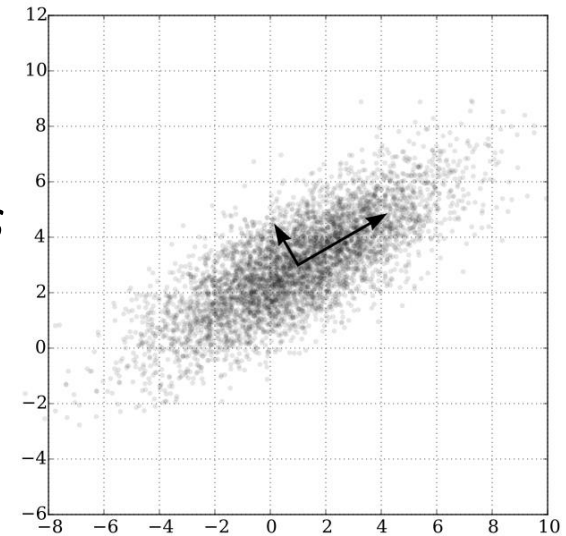
- ❑ PCA: A statistical procedure that uses an orthogonal transformation to convert a set of observations of possibly correlated variables into a set of values of linearly uncorrelated variables called *principal components*
- ❑ The original data are projected onto a much smaller space, resulting in dimensionality reduction
- ❑ Method: Find the eigenvectors of the covariance matrix, and these eigenvectors define the new space



Ball travels in a straight line. Data from three cameras contain much redundancy

# Principal Component Analysis (Method)

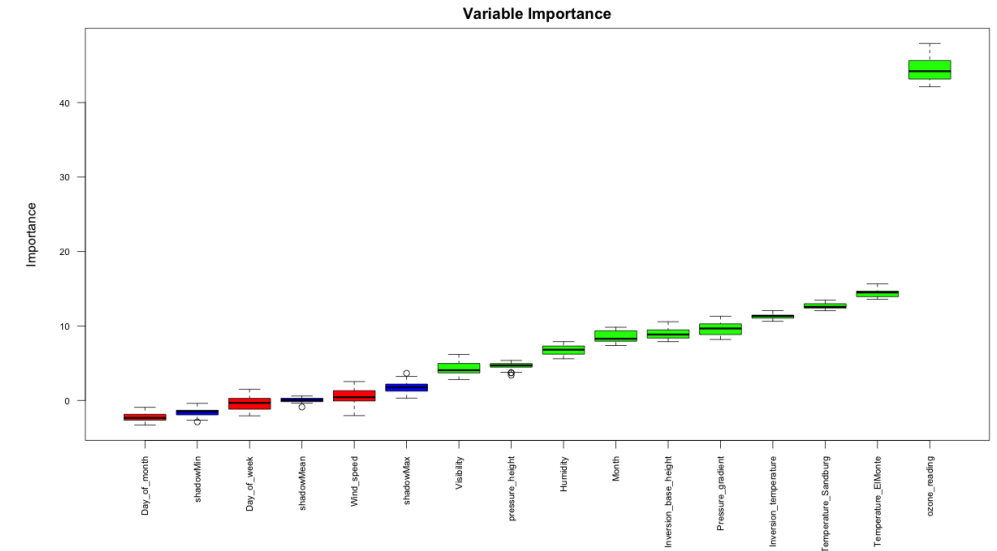
- Given  $N$  data vectors from  $n$ -dimensions, find  $k \leq n$  orthogonal vectors (*principal components*) best used to represent data
  - Normalize input data: Each attribute falls within the same range
  - Compute  $k$  orthonormal (unit) vectors, i.e., *principal components*
  - Each input data (vector) is a linear combination of the  $k$  principal component vectors
  - The principal components are sorted in order of decreasing “significance” or strength
  - Since the components are sorted, the size of the data can be reduced by eliminating the *weak components*, i.e., those with low variance (i.e., using the strongest principal components, to reconstruct a good approximation of the original data)
- Works for numeric data only



Ack. Wikipedia: Principal Component Analysis

# Attribute Subset Selection

- Another way to reduce dimensionality of data
- Redundant attributes
  - Duplicate much or all of the information contained in one or more other attributes
    - E.g., purchase price of a product and the amount of sales tax paid
- Irrelevant attributes
  - Contain no information that is useful for the data mining task at hand
    - Ex. A student's ID is often irrelevant to the task of predicting his/her GPA



# Heuristic Search in Attribute Selection

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- ❑ There are  $2^d$  possible attribute combinations of  $d$  attributes
- ❑ Typical heuristic attribute selection methods:
  - ❑ Best single attribute under the attribute independence assumption: choose by significance tests
  - ❑ Best step-wise feature selection:
    - ❑ The best single-attribute is picked first
    - ❑ Then next best attribute condition to the first, ...
  - ❑ Step-wise attribute elimination:
    - ❑ Repeatedly eliminate the worst attribute
  - ❑ Best combined attribute selection and elimination
  - ❑ Optimal branch and bound:
    - ❑ Use attribute elimination and backtracking

# Attribute Creation (Feature Generation)

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- ❑ Create new attributes (features) that can capture the important information in a data set more effectively than the original ones
- ❑ Three general methodologies
  - ❑ Attribute extraction
    - ❑ Domain-specific
  - ❑ Mapping data to new space (see: data reduction)
    - ❑ E.g., Fourier transformation, wavelet transformation, manifold approaches (not covered)
  - ❑ Attribute construction
    - ❑ Combining features (see: discriminative frequent patterns in Chapter on “Advanced Classification”)
    - ❑ Data discretization



# Summary

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- ❑ **Data quality:** accuracy, completeness, consistency, timeliness, believability, interpretability
- ❑ **Data cleaning:** e.g. missing/noisy values, outliers
- ❑ **Data integration** from multiple sources:
  - ❑ Entity identification problem; Remove redundancies; Detect inconsistencies
- ❑ **Data reduction, data transformation and data discretization**
  - ❑ Numerosity reduction; Data compression
  - ❑ Normalization; Concept hierarchy generation
- ❑ **Dimensionality reduction**

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